

5.3.6.1 Current consumption in Run mode

Table 24. Typical and maximum current consumption in Run mode

Specified by design - not tested in production, unless otherwise stated.

Clocked by HSI at 144 MHz or HSIDIV3 at 48MHz if not otherwise specified.

Symbol	Parameter	Conditions	f _{HCLK} (MHz)	Typ	Max LDO		Unit
					T _J = 30°C	T _J = 140°C	
I _{DD(Run)} ⁽¹⁾	Supply current in Run mode with all peripheral clocks disabled	Code in Flash, ICACHE 2-way	144	11.4	12.00	24.00	mA
			48	4.1	4.20	17.50	
		Code in Flash, ICACHE 1-way	144	10.5	11.00	23.00	
			48	3.8	4.00	17.50	
		Code in Flash, ICACHE OFF	144	11.3	12.00	23.50	
			48	4.6	5.00	18.00	
	Supply current in Run mode with all peripheral clocks enabled	Code in SRAM2, ICACHE OFF, FLASH ON	144	10.5	11.00	23.50	
			48	3.8	4.00	17.50	
		Code in flash, ICACHE 2-way	144 ⁽²⁾	35.00	37.00	47.00	
		Code in flash, ICACHE OFF	144 ⁽²⁾	35.00	37.00	47.00	

1. Measurements done with prefetch enabled.

2. Clocked by PSI at 144 MHz with HSE at 16 MHz in bypass mode.

Table 25. Typical current consumption in Run mode with CoreMark running from flash memory and SRAM

Specified by design - not tested in production, unless otherwise stated.

Symbol	Parameter	Conditions	SYSCLK source	(MHz)	Typ	Unit	Typ	Unit
		Peripheral						
I _{DD(Run)} ⁽¹⁾	Supply current in Run mode	All peripherals disabled, Instruction cache 2-WAY, Prefetch ON	PSI ⁽²⁾	144	11.00	mA	78.00	μA/ MHz
		All peripherals disabled, Instruction cache 1-WAY, Prefetch ON			9.95		69.50	
		All peripherals disabled, Instruction cache OFF, Prefetch ON			11.50		80.00	
		All peripherals disabled, Instruction cache OFF, Prefetch OFF			10.00		69.50	
		All peripherals disabled, SRAM2, Instruction cache 2-WAY, FLASH ON			11.00		75.00	
		All peripherals disabled, SRAM2, Instruction cache 1-WAY, FLASH ON			9.60		66.50	
		All peripherals disabled, SRAM2, Instruction cache OFF, FLASH ON			10.50		71.50	

1. Measures done with prefetch enabled.

2. Clocked by PSI 144 MHz with HSE at 16 MHz on bypass mode.